

MODULE-4

DYNAMIC PROGRAMMING, BRANCH AND

BOUND & BACKTRACKING

DYNAMIC PROGRAMMING (recursive)

principle of optimality:-

It states that an optimal sequence of decision has the property that whatever the initial state and decisions are, the remaining decisions must constitute the optimal decision sequence with regard to the state resulting from 1st decision.

characteristics of Dynamic pro:-

1. first Optimal substructure.
2. Overlapping Subproblems.

STEPS OF DYNAMIC PROGRAMMING APPROACH:-

1. characterise the structure of an optimal solution.
2. Recursively define the value of an optimal solution.
3. Compute the value of an optimal solution typically in a bottom up fashion.
4. Construct an optimal solution from the computed information.

Applications:-

1. matrix chain multiplication
2. All pair shortest path problem.

Imp

MATRIX CHAIN MULTIPLICATION

problem stated that

"Given a chain $\langle A_1, A_2, \dots, A_n \rangle$ of n matrices where for $i=1, 2, \dots, n$ matrix A_i has dimension $P_{i-1} \times P_i$ fully parenthesise the product $A_1, A_2, \dots, A_n$ in a way that minimise the no. of scalar multiplication."

multiplication of matrix:-

- If dimension of matrix A is $P \times Q$ and B is $Q \times R$ then resultant matrix is $P \times R$.
- Computational cost = $P \times Q \times R$.

Q. Consider the problem of a chain A_1, A_2, A_3 of 3 matrices. Suppose A_1 is a 10×100 matrix. A_2 is 100×5 matrix and A_3 is 5×50 matrix. find

ans:- Possible ways to multiply matrices = ^{computation} Cost.
 $(A_1 A_2) A_3$ and $A_1 (A_2 A_3)$

$$(A_1 A_2) A_3 = A_1 = 10 \times 100$$

$$A_2 = 100 \times 5$$

$$A_3 = 5 \times 50$$

$$\Rightarrow A_1 A_2 = 10 \times 5$$

$$A_2 A_3 = 100 \times 50$$

Let, $(A_1 A_2) = C$ then

$$\text{Computation Cost} = P \times Q \times R \Rightarrow (10 \times 100 \times 5)$$

$$\text{Size of } C \cdot A_3 = (10 \times 5 \times 50)$$

$$\Downarrow$$

$$10 \times 5$$

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$$\text{then } (A_1 A_2) A_3 = (10 \times 100 \times 5) + (10 \times 5 \times 50) \\ = 7500.$$

$A_1(A_2 A_3) \Rightarrow A_2 A_3 = (100 \times 5 \times 50)$ It is denoted as C . Then size of $C = 100 \times 50$.

$\Rightarrow$ Size of $A_2 \cdot A_3 = 100 \times 50 \Rightarrow C$.

then Computational Cost = $100 \times 5 \times 50$

$C = 100 \times 50$ then $A_1 C = 10 \times 100 \times 50$

$$\text{then } A_1(A_2 A_3) = (100 \times 5 \times 50) + (10 \times 100 \times 50) \\ = 15000$$

Algorithm:-

- Table S is used to construct an optimal Solution.
- Table M is used to store $M[i, j]$ cost.
- Table M uses only the main diagonal and the upper triangle and a stable uses only upper triangle.

MATRIX CHAIN ORDER (P)

$n = P.length - 1$

Let $m[1 \dots n, 1 \dots n]$ and $s[1 \dots n-1, 2 \dots n]$ be new tables.

for $i = 1$ to n

$m[i, i] = 0$

```

for i=2 to n
    for j=i+1 to n
        m[i,j] = ∞
        for k=i to j-1
            q = m[i,k] + m[k+1,j] + Pk · Pj
            if q < m[i,j]
                m[i,j] = q
                s[i,j] = k
PRINT - OPTIMAL - PARENS(s, i, j)
if i == j
    print "A";
else print ("PRINT - OPTIMAL - PARENS
            (s, i, s[i,j])
            print - OPTIMAL - PARENS(s, s[i,j]+1, j)
            print ")
    
```

- Q. Given a chain of 4 matrices A_1, A_2, A_3 and A_4 with dimensions $5 \times 4, 4 \times 6, 6 \times 2, 2 \times 7$ respectively. Using dynamic programming find minimum no. of scalar multiplications and also write the ^{optimal} multiplication order.

$$((A_1(A_2 A_3))A_4)$$

$$((A_1(A_2 A_3))A_4)$$

ans:-

m	1	2	3	4	5	1	2	3	4
1	0	120	88	156	1		1	1	3
2		0	48	104	2			2	3
3			0	84	3				3
4				0	4				

$A_1 \quad A_2 \quad A_3 \quad A_4$

$5 \times 4 \quad 4 \times 6 \quad 6 \times 2 \quad 2 \times 7$

$$A_1 A_2 \Rightarrow A_1 = 5 \times 4$$

$$A_2 = 4 \times 6$$

$$\text{Size of } A_1 A_2 = 5 \times 6$$

$$\text{Cost of } A_1 A_2 = 5 \times 4 \times 6 = 120 //$$

$$A_2 A_3 \Rightarrow 4 \times 6 \times 2 = 48 //$$

$$A_3 A_4 \Rightarrow 6 \times 2 \times 7 = 84 //$$

possible multiplication:- $A_1 A_2 A_3, A_2 A_3 A_4$

$A_1 A_2 A_3$ can be multiplied in 2 ways.

I. 1. $A_1 (A_2 A_3)$

2. $(A_1 A_2) A_3$

$$A_1 (A_2 A_3)$$

$$5 \times 4 \quad 4 \times 6 \quad 6 \times 2$$

$$4 \times 6 \times 2 = 48$$

$$\text{Size of } A_2 A_3 = 4 \times 2$$

$$5 \times 4 \quad 4 \times 2$$

$$5 \times 4 \times 2 = 40$$

$$\text{Then total cost} = 40 + 48 = 88 //$$

$$(A_1 (A_2 A_3))$$

$\Rightarrow$ Then take A_1

if $(A_2 A_3 A_4)$

Then take A_2

if $(A_3 A_4) A_5$

Then take A_4

$$2. (A_1 A_2) A_3$$

$$5 \times 4 \quad 4 \times 6 \quad 6 \times 2$$

$$\text{Cost} = 5 \times 4 \times 6 = 120 //$$

$$\text{Size} = 5 \times 6 \quad 6 \times 2$$

$$\text{Cost} = 5 \times 6 \times 2 = 60 //$$

$$\text{Total Cost} = 120 + 60 = 180 //$$

Here $A_1 (A_2 A_3) < (A_1 A_2) A_3$.

Then write in column of $A_1 A_3 \Rightarrow 153 //$

II. $A_2 A_3 A_4$

$$1. A_2 (A_3 A_4)$$

$$4 \times 6 \quad 6 \times 2 \quad 2 \times 7$$

$$\text{Cost} = 6 \times 2 \times 7 = 84 //$$

$$\text{Size} = 6 \times 7$$

$$4 \times 6 \quad 6 \times 7$$

$$\text{Cost} = 4 \times 6 \times 7 = 168 //$$

$$\text{Total Cost} = 252 //$$

$$2. (A_2 (A_3 A_4))$$

$$4 \times 6 \quad 6 \times 2 \quad 2 \times 7$$

$$\text{Cost} = 4 \times 6 \times 2 = 48 //$$

$$\text{Size} = 4 \times 2$$

$$4 \times 2$$

$$2 \times 7$$

$$\text{Cost} = 4 \times 2 \times 7$$

$$= 56$$

$$\text{Total Cost} =$$

$$48 + 56$$

$$= 104 //$$

Here $(A_2 A_3) A_4 \leq A_2 (A_3 A_4) //$

possible multiplications: $-(A_1 A_2 A_3) A_4$

$(A_1 A_2 A_3) A_4$

8×4

1. 5×4 4×6 6×2 2×7

Cost = 88

Size = 5×2

Cost = $5 \times 2 \times 7 = 70$

Total cost = 158 // Smallest one,

2. $A_1 (A_2 A_3 A_4)$ Then take as 3.

Cost = 104

Size = 4×7

Cost = $5 \times 4 \times 7 = 140$

Total cost = 244 //

3. $(A_1 A_2) (A_3 A_4)$

~~$A_1 (A_2 A_3 A_4)$~~

120

(5×6)

84 (6×7)

210 = $5 \times 6 \times 7$

total = $210 + 120 + 84$

= 414 //

Then Optimal Order =

$(A_1 (A_2 A_3)) A_4$

Q. 11.2 Given a chain of 5 matrices A_1, A_2, A_3, A_4, A_5 with dimensions $4 \times 10, 10 \times 3, 3 \times 12, 12 \times 20, 20 \times 7$ respectively. Using dynamic programming, find minimum no. of scalar multiplication also write its optimal order.

ans:- m

		1	2	3	4	5	5	1	2	3	4	5
$m(1,1)$	1	0	120	264	1080	1344	1		21	2	2	2
$m(2,2)$	2		0	360	1320	1350	2			2	2	2
$m(3,3)$	3			0	720	1140	3				3	4
$m(4,4)$	4				0	1680	4					4
$m(5,5)$	5					0	5					

$A_1 \quad A_2 \quad A_3 \quad A_4 \quad A_5$
 $4 \times 10 \quad 10 \times 3 \quad 3 \times 12 \quad 12 \times 20 \quad 20 \times 7$

$A_1 A_2 = 4 \times 10 \quad 10 \times 3$
 $\text{cost} = 4 \times 10 \times 3 = 120$
 $\text{Size} = 4 \times 3$

$A_2 A_3$
 $10 \times 3 \quad 3 \times 12$
 $\text{cost} = 10 \times 3 \times 12 = 360$

$\text{Size} = 10 \times 12$

$A_3 A_4$
 $3 \times 12 \quad 12 \times 20$
 $\text{cost} = 3 \times 12 \times 20 = 720$
 $\text{Size} = 3 \times 20$

$A_4 A_5$
 $12 \times 20 \quad 20 \times 7$
 $\text{cost} = 12 \times 20 \times 7 = 1680$
 $\text{Size} = 12 \times 7$

possible multiplications with 3 matrices :-

1. $A_1 A_2 A_3$
2. $A_2 A_3 A_4$
3. $A_3 A_4 A_5$

1. $A_1 A_2 A_3$ - can be multiplied in 2 ways.

1. $A_1 (A_2 A_3)$

2. $(A_1 A_2) A_3$

$$\begin{array}{ccc} 4 \times 10 & 10 \times 3 & 3 \times 12 \\ & \underbrace{10 \times 3 \times 12 = 360} & \\ & \text{Size} = 10 \times 12 & \end{array}$$

$$\begin{array}{ccc} 4 \times 10 & 10 \times 3 & 3 \times 12 \\ & \underbrace{4 \times 10 \times 3 = 120} & \\ & \text{Size} = 4 \times 3 & \end{array}$$

$$4 \times 10 \times 12 = 480$$

$$4 \times 3 \times 3 \times 12$$

$$\text{Total cost} = 480 + 360 = 840 //$$

$$\text{Cost} = 4 \times 3 \times 12$$

$$= 144 //$$

$$\text{Total Cost} = 264 //$$

Here $A_1 (A_2 A_3) > (A_1 A_2) A_3$.

2. $A_2 A_3 A_4$ - can be multiplied in 2 ways.

1. $A_2 (A_3 A_4)$

2. $(A_2 A_3) A_4$

$$\begin{array}{ccc} 10 \times 3 & 3 \times 12 & 12 \times 20 \\ & \underbrace{\text{Cost} = 3 \times 12 \times 20 = 720} & \\ & \text{Size} = 3 \times 20 & \end{array}$$

$$\begin{array}{ccc} 10 \times 3 & 3 \times 12 & 12 \times 20 \\ & \underbrace{10 \times 3 \times 12 = 360} & \end{array}$$

$$\text{Cost} = 10 \times 3 \times 20 = 600$$

$$S = 10 \times 12$$

$$\text{Total} = 1320 //$$

$$\text{Cost} = 10 \times 12 \times 20$$

$$\text{Total} = 2760 //$$

Here $A_2 (A_3 A_4) < (A_2 A_3) A_4$.

3. $A_3 A_4 A_5$ can be multiplied in 2 ways.

1. $A_3 (A_4 A_5)$

2. $(A_3 A_4) A_5$

$$\begin{array}{ccc} 3 \times 12 & 12 \times 20 & 20 \times 7 \\ & \underbrace{12 \times 20 \times 7 = 1680} & \\ & \underbrace{12 \times 7} & \end{array}$$

$$\begin{array}{ccc} 3 \times 12 & 12 \times 20 & 20 \times 7 \\ & \underbrace{3 \times 12 \times 20 = 720} & \\ & \underbrace{3 \times 20} & \end{array}$$

$$3 \times 12 \times 7 = 252$$

$$3 \times 20 \times 7 = 420$$

$$\text{Total} = 1932 //$$

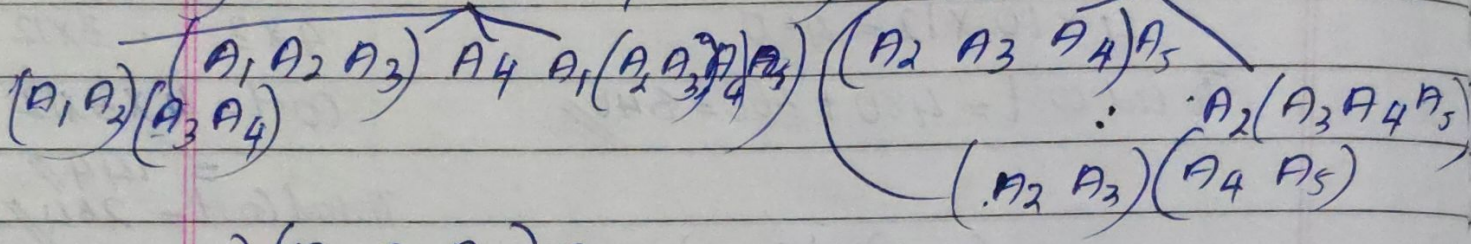
$$\text{Total} = 1140 //$$

possible multiplication with 4 matrices :-

1. $A_1(A_2 A_3 A_4 A_5)$ 2. $(A_1 A_2 A_3 A_4) A_5$
 3. $A_1((A_2 A_3)(A_4 A_5))$ 4. $((A_1 A_2)(A_3 A_4)) A_5$

1. $(A_1 A_2 A_3) A_4$ 2. $A_1(A_2 A_3 A_4)$
 3. $A_1 A_2 A_3 A_4 A_5$

- 1) $A_1 A_2 A_3 A_4$ 2) $A_2 A_3 A_4 A_5$



- 1) $(A_1 A_2 A_3) A_4$
 $4 \times 10 \quad 10 \times 3 \quad 3 \times 12 \quad 12 \times 20$

Cost = 264

Size = 4×12

Cost = $4 \times 12 \times 20 = 960$

Total = 1224

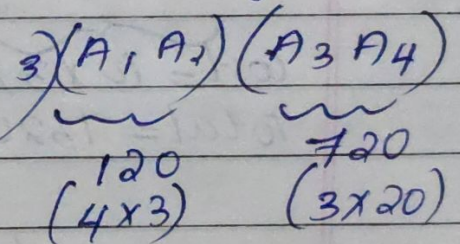
- 2) $A_1(A_2 A_3 A_4)$
 $4 \times 10 \quad 10 \times 3 \quad 3 \times 12 \quad 12 \times 20$

Cost = 1320

Size = 10×20

Cost = $4 \times 10 \times 20 = 800$

Total = 2120



120
(4x3)

720
(3x20)

Cost = $4 \times 3 \times 20 = 240$

Total = $120 + 720 +$

240

= 1080

Continuation in back side :-

Continuation of matrix chain multiplication problem

2) $A_2 \cdot A_3 \cdot A_4 \cdot A_5$

$\bullet A_1 (A_2 A_3 A_4) = 4 \times 10 \ 10 \times 3$

1) $(A_2 A_3 A_4) A_5$

$3 \times 12 \ 12 \times 20$

$10 \times 3 \quad 3 \times 12 \quad 3 \times 12 \quad 20 \times 7$
 12×20

1320

$2120 //$

1320
 (10×20)

3) $(A_2 A_3) (A_4 A_5)$

$\text{cost} = 10 \times 20 \times 7 = 1400$

$360 \quad 1680$

$\text{Total} = 1320 + 1400 = 2720 //$

$(10 \times 12) \quad (12 \times 7)$

2) $A_2 (A_3 A_4 A_5)$

$10 \times 3 \quad 3 \times 12 \quad 12 \times 20 \quad 20 \times 7$

~~2040~~

$\text{cost} = 10 \times 12 \times 7$
 $= 840$

1140
 (3×7)

$\text{Total} = 840 +$

$\text{cost} = 10 \times 3 \times 7 = 210$

$360 +$

1680

$\text{Total} = 1350 //$

$2880 //$

possible multiplication with 5 matrices:-

1. $A_1(A_2 A_3 A_4 A_5)$
2. $(A_1 A_2 A_3 A_4) A_5$

I. $A_1(A_2 A_3 A_4 A_5)$ can be multiplied as

1. $A_1(A_2 A_3 A_4 A_5)$
2. $(A_1 A_2 A_3 A_4) A_5$

$$4 \times 10 \quad 10 \times 3 \quad 3 \times 12 \quad 12 \times 20 \quad 20 \times 7$$

$$\text{Cost} = 10 \times 3 \times 12 \times 20 \times 7$$

$$= 1350$$

$$\text{Size} = 10 \times 7$$

$$\text{Cost} = 4 \times 10 \times 7 = 280$$

$$(4 \times 7)$$

$$\text{Total} = 1350 + 280 = 1630$$

$$3) A_1 A_2 (A_3 A_4 A_5)$$

$$= \underbrace{120}_{A_1 A_2} \underbrace{1140}_{A_3 A_4 A_5}$$

$$1344$$

$$4 \times 10 \quad 10 \times 3 \quad 3 \times 12 \quad 12 \times 20 \quad 20 \times 7$$

$$C = 1080$$

$$\text{Size} = 4 \times 20$$

$$\text{Cost} = 4 \times 20 \times 7$$

$$= 560$$

$$(4 \times 7)$$

Then optimal

order =

$$(A_1 (A_2 A_3) (A_4 A_5))$$

∴ Minimum

$$\text{Scalar multiplication} = 1344$$

$$\text{Total} = 560 + 1080$$

$$= 1640$$

~~ALL PAIR SHORTEST PATH ALGORITHM:-~~

FLOYD-WORSHOLL ALGORITHM:-

Let V = no. of vertices in graph.

Let $dist$ = $V \times V$ array of minimum distances.

for each Vertex v

$dist[v][v] = 0$

for each edge (u, v)

$dist[u][v] = weight(u, v)$

for k from 1 to v

for i from 1 to v

for j from 1 to v

if $(dist[i][j] > dist[i][k] + dist[k][j])$

$dist[i][j] = \min\{dist[i][j], dist[i][k] + dist[k][j]\}$

end if.

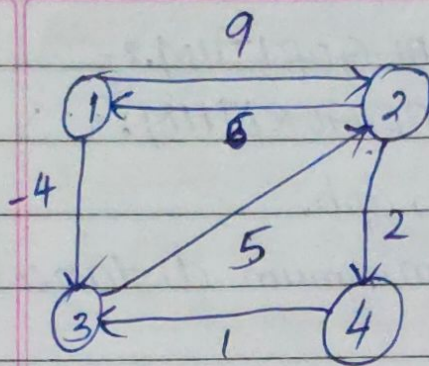
Analysis :-

3 loops = $O(n^3)$.

$n \rightarrow$ no. of nodes in the given graph.

Space Complexity = $O(n^2)$

Q. Using Floyd-warshall algo find shortest path distance with every pair of vertices.



Adjacency matrix:-

	1	2	3	4
1	0	9	-4	∞
2	6	0	∞	2
3	∞	5	0	∞
4	∞	∞	1	0

$D_1 =$	1	2	3	4
1	0	9	-4	∞
2	6	0	2	2
3	∞		0	
4	∞			0

$$D^1(2,3) = \min_i \{ D^0(2,3), D^0(2,i) + D^0(i,3) \}$$

$$= \min \{ \infty, 6 + (-4) \} = \min \{ \infty, 2 \}$$

$$D^1(2,4) = \min_j \{ D^0(2,4), D^0(2,i) + D^0(i,j) \}$$

$$= \min \{ 2, 6 + \infty \}$$

$$D^1(3,2) = \min_i \{ D^0(3,2), D^0(3,i) + D^0(i,2) \}$$

$$= \min \{ 5, (\infty + 9) \}$$

$$= \min \{ 5, \infty \} = 5 //$$

$$D'(3,4) = \min \{ D^0(3,4), D^0(3,1) + D^0(1,4) \}$$

$$= \min \{ \infty, \infty + \infty \}$$

$$= \min \{ \infty, \infty \}$$

$$= \infty$$

$$D'(4,2) = \min \{ D^0(4,2), D^0(4,1) + D^0(1,2) \}$$

$$= \min \{ \infty, \infty + 9 \}$$

$$= \min \{ \infty \} = \infty$$

$$D'(4,3) = \min \{ D^0(4,3), D^0(4,1) + D^0(1,3) \}$$

$$= \min \{ 1, \infty + -4 \}$$

$$= 1$$

	1	2	3	4
1	0	9	-4	∞
2	6	0	2	2
3	∞	5	0	∞
4	∞	∞	1	0

	1	2	3	4
1	0	9	-4	11
2	6	0	2	2
3	11	5	0	7
4	∞	∞	1	0

$$D^2(1,3) = \min\{D^1(1,3), D^1(1,2) + D^1(2,3)\}$$

$$= \min\{-4, (9+2)\}$$

$$= \min\{-4, 11\}$$

$$D^2(1,4) = \min\{D^1(1,4), D^1(1,2) + D^1(2,4)\}$$

$$= \min\{\infty, (9+2)\}$$

$$= \min\{\infty, 11\}$$

$$D^2(3,1) = \min\{D^1(3,1), D^1(3,2) + D^1(2,1)\}$$

$$= \min\{-4, (5+6)\}$$

$$D^2(3,4) = \min\{D^1(3,4), D^1(3,2) + D^1(2,4)\}$$

$$= \min\{\infty, (5+2)\}$$

$$D^2(4,1) = \min\{D^1(4,1), D^1(4,2) + D^1(2,1)\}$$

$$= \min\{\infty, (\infty+6)\}$$

$$D^2(4,3) = \min\{D^1(4,3), D^1(4,2) + D^1(2,3)\}$$

$$= \min\{1, (\infty+2)\}$$

$$= 1$$

Ans

	1	2	3	4
D^3				
1			-4	
2			2	
3	11	5	0	7
4			1	

$$\begin{aligned}
 D^3(1,1) &= \min \{ D^2(1,1), D^2(1,3) + D^2(3,1) \} \\
 &= \min \{ 0, (-4 + 11) \} \\
 &= \min \{ 0, 7 \}
 \end{aligned}$$

$= 0 //$

$$\begin{aligned}
 D^3(1,2) &= \min \{ D^2(1,2), D^2(1,3) + D^2(3,2) \} \\
 &= \min \{ 9, (-4 + 5) \} \\
 &= \min \{ 9, 1 \}
 \end{aligned}$$

$= 1 //$

$$\begin{aligned}
 D^3(1,4) &= \min \{ D^2(1,4), D^2(1,3) + D^2(3,4) \} \\
 &= \min \{ 11, (-4 + 7) \} \\
 &= \min \{ 11, 3 \}
 \end{aligned}$$

$= 3 //$

$$\begin{aligned}
 D^3(2,1) &= \min \{ D^2(2,1), D^2(2,3) + D^2(3,1) \} \\
 &= \min \{ 6, (2 + 11) \}
 \end{aligned}$$

$= 6 //$

$$\begin{aligned}
 D^3(2,2) &= \min \{ D^2(2,2), D^2(2,3) + D^2(3,2) \} \\
 &= \min \{ 0, (2 + 5) \} = 0 //
 \end{aligned}$$

$$D^3(2,4) = \min \left\{ D^2(2,4), D^2(2,3) + D^2(3,4) \right\}$$

$$= \min \left\{ 2, (2+7) \right\}$$

$$= 2 //$$

$$D^3(4,1) = \min \left\{ D^2(4,1), D^2(4,3) + D^2(3,1) \right\}$$

$$= \min \left\{ \infty, (1+11) \right\}$$

$$= 12 //$$

$$D^3(4,2) = \min \left\{ D^2(4,2), D^2(4,3) + D^2(3,2) \right\}$$

$$= \min \left\{ \infty, (1+5) \right\}$$

$$= \min \left\{ \infty, 6 \right\}$$

$$= 6 //$$

$$D^3(4,4) = \min \left\{ D^2(4,4), D^2(4,3) + D^2(3,4) \right\}$$

$$= \min \left\{ 0, (1+7) \right\}$$

$$= 0 //$$

Then, $D^3 \Rightarrow$

	1	2	3	4
1	0	1	-4	3
2	6	0	2	2
3	11	5	0	7
4	12	6	1	0

Then Compute D^4 by placing 4th row and 4th column take as same, and find remaining.

	1	2	3	4
$D^4 = 1$				3
2				2
3				7
4	12	6	1	0

$$D^4(1,1) = \min\{D^3(1,1), D^3(1,4) + D^3(4,1)\}$$

$$= \min\{0, (3 + 0)\} = 0 //$$

$$D^4(1,2) = \min\{D^3(1,2), D^3(1,4) + D^3(4,2)\}$$

$$= \min\{1, (3 + 6)\} = 1 //$$

$$D^4(1,3) = \min\{D^3(1,3), D^3(1,4) + D^3(4,3)\}$$

$$= \min\{-4, (3 + 1)\} = -4 //$$

$$D^4(2,1) = \min\{D^3(2,1), D^3(2,4) + D^3(4,1)\}$$

$$= \min\{6, (2 + 12)\} = 6 //$$

$$D^4(2,2) = \min\{D^3(2,2), D^3(2,4) + D^3(4,2)\}$$

$$= \min\{0, (2 + 6)\} = 0 //$$

$$D^4(2,3) = \min\{D^3(2,3), D^3(2,4) + D^3(4,3)\}$$

$$= \min\{2, (2 + 1)\} = 2 //$$

$$D^4(3,1) = \min\{D^3(3,1), D^3(3,4) + D^3(4,1)\}$$

$$= \min\{11, (7 + 0)\} = 11 //$$

$$D^4(3,2) = \min\{D^3(3,2), D^3(3,4) + D^3(4,2)\}$$

$$= \min\{5, (7 + 6)\} = 5 //$$

$$D^4(3,3) = \min\{D^3(3,3), D^3(3,4) + D^3(4,3)\}$$

$$= \min\{0, (7 + 1)\} = 0 //$$

then the resultant matrix will be:-

$D_4 =$

	1	2	3	4
1	0	1	-4	3
2	6	0	2	2
3	11	5	0	7
4	12	6	1	0

BACKTRACKING

Algorithm Backtrack(k)

```

{
  for each ( $x(k) \in T[x(1), x(2), \dots, x(k-1)]$ )
  {
    if ( $B_k(x(1), x(2), \dots, x(k)) \neq 0$ )
    {
      if
         $x(1), x(2), \dots, x(k)$  is a path an answer
        node.
      write ( $x(1:k)$ )
    }
    if ( $k < n$ ) Backtrack( $k+1$ )
  }
}

```


	1	2	3	4	#diagonal = row-column
1	0	-1	-2	-3	$(1,1) = 1-1=0$
2	1	0	-1	-2	$(1,2) = 1-2 = -1 = +ve$
3	2	1	0	-1	$(1,3) = -2 = +ve$
4	3	2	1	0	$(1,4) = -3 = +ve$

N QUEENS PROBLEM

ALGORITHM:-

```

Algorithm NQueens(k,n)
{
  for i=1 to n do
  {
    if place(k,i) then
    {
      x[k]=i
      if (k==n) write x[1:n]
      else NQueen(k+1,n)
    }
  }
}

```

```

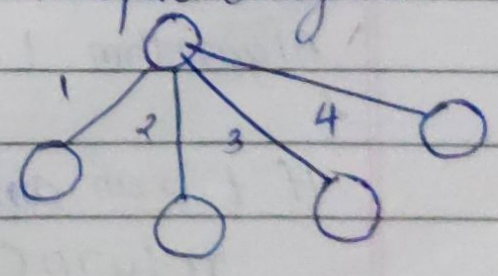
Algorithm place(k,i)
{
  for j=1 to k-1
  {
    if x[j]==i or
    (Abs(x[j]-i) == Abs(j-k))
  }
  return false
}
Return true
}

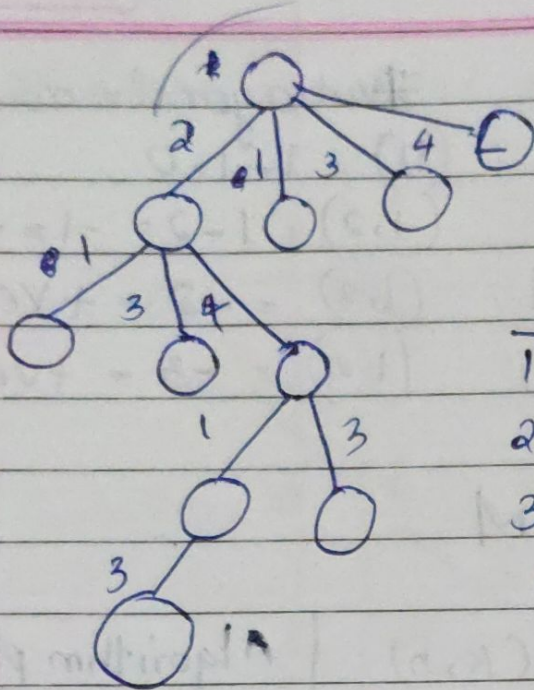
```

problem of 4 Queens :-

	1	2	3	4
1		Q ₁		
2				Q ₂
3	Q ₃			
4			Q ₄	

State Space diagram :-

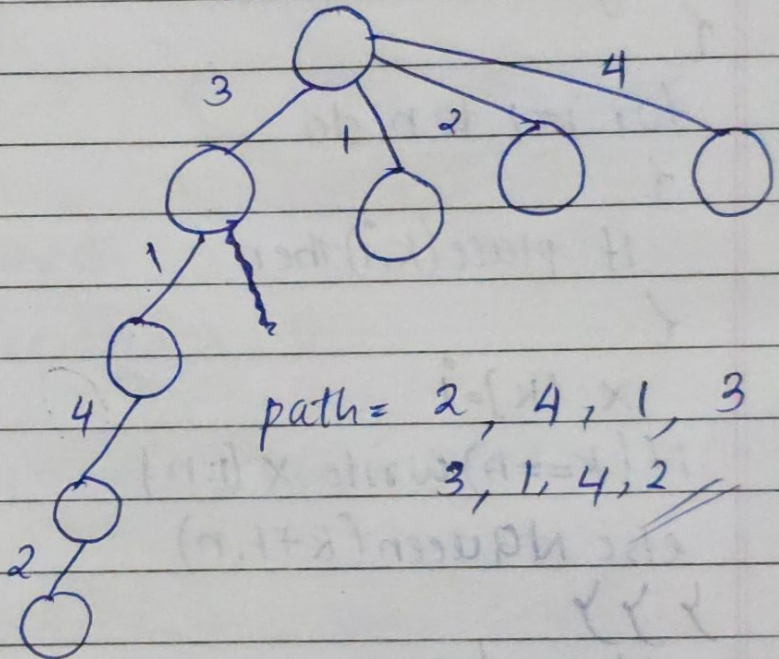




Solution 2 :-

	1	2	3	4
1			Q ₁	
2	Q ₂			
3				Q ₃
4		Q ₄		

path = 3, 1, 4, 2



path = 2, 4, 1, 3 or
3, 1, 4, 2

BRANCH AND BOUND

Algorithm LC search(t)

{

If t is an answer node

return C

$R = t$

Initialise the list of live node to be empty.

repeat

{

for each child x of t do {

If x is an answer node then output the path from x to t & return

add (x)

$x \rightarrow \text{parent} = E.$

}

If there are no more live nodes then

{

write "no answer node"

return

}

$E = \text{least}()$

} until (false)

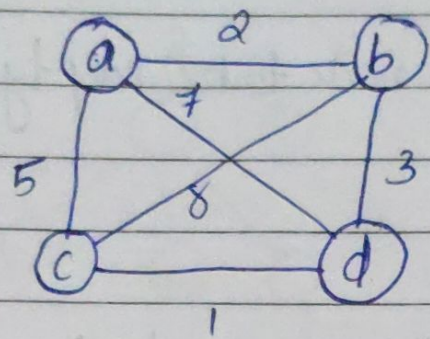
}

TSP (Travelling Salesman Problem)

- It is implemented by using Branch and Bound Algorithm.

Q. perform Tsp for the graph.

updating elements by choosing smallest one and then subtract all elements by it.



Adjacency matrix:-

	a	b	c	d
a	∞	2	5	7
b	2	∞	8	3
c	5	8	∞	1
d	7	3	1	∞

- Implementing self node as infinity.

- 1. Row Reduction
- 2. Column Reduction

Smallest element
2
then - all

	a	b	c	d
a	∞	0	+3	+5
b	0	∞	6	1
c	4	7	∞	0
d	6	2	0	∞

Then perform Column Reduction:-

$M_1 \Rightarrow$

	a	b	c	d
a	∞	0	3	5
b	0	∞	6	1
c	4	7	∞	0
d	6	2	0	∞

Then we want to find total no. of reductions:-

$2 + 2 + 1 + 1 = 6 + 0 = 6$

row reduction

then cost = 6

Column reduction

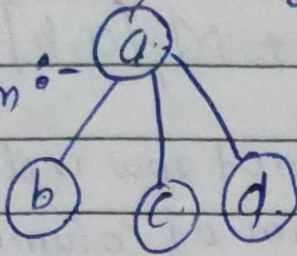
lowest element in 1st

row

lowest element in 2nd

Starting node

state diagram



	B			
	a	b	c	d
a	∞	0	3	5
b	0	∞	6	1
c	4	7	∞	0
d	6	2	0	∞

↓

	a	b	c	d
a	∞	∞	∞	∞
b	∞	∞	6	1
c	4	∞	∞	0
d	6	∞	0	∞

Then perform Row Reduction $\Rightarrow$

①

	a	b	c	d
a	∞	∞	∞	∞
b	∞	∞	5	0
c	4	∞	∞	0
d	6	∞	0	∞

Column Reduction $\Rightarrow$

	a	b	c	d
a	∞	∞	∞	∞
b	∞	∞	5	0
c	0	∞	∞	0
d	2	∞	0	∞

Row Reduction = $0 + 1 + 0 + 0 = 1 //$

~~Cost reduced = Cost of node + $M, [a, b] + x \leq M_b$~~

C

∞	∞	∞	∞
0	∞	∞	1
∞	7	∞	0
6	2	∞	∞

Column Reduction = $4 + 0 + 0 + 0 = 4 //$

Then $x = 1 + 4 = 5 //$

$$\text{cost reduced} = \text{Cost of node 1} + M_1[a, b] + \gamma$$

$$= 6 + 0 + 5$$

$$= 11$$

Sum of row reduction +
Sum of column reduction

For Finding C

$$\begin{bmatrix} \infty & \infty & \infty & \infty \\ 0 & \infty & \infty & 1 \\ \infty & 7 & \infty & 0 \\ 6 & 2 & \infty & \infty \end{bmatrix}$$

i) Row Reduction

$$\begin{bmatrix} \infty & \infty & \infty & \infty \\ 0 & \infty & \infty & 1 \\ \infty & 7 & \infty & 0 \\ 4 & 0 & \infty & \infty \end{bmatrix}$$

$$\text{no. of steps} = 0 + 0 + 0 + 2 = 2$$

ii) Column Reduction

$$M_3 = \begin{bmatrix} \infty & \infty & \infty & \infty \\ 0 & \infty & \infty & 1 \\ \infty & 7 & \infty & 0 \\ 4 & 0 & \infty & \infty \end{bmatrix}$$

$$\text{no. of steps} = 0 + 0 + 0 + 0 = 0$$

$$\text{Then cost} = \text{Cost of Node 1} + M_1^c[a, b] + \gamma$$

$$= 6 + 3 + 2$$

$$= 11$$

$$D = \begin{bmatrix} \infty & \infty & \infty & \infty \\ 0 & \infty & 6 & \infty \\ 4 & 7 & \infty & \infty \\ \infty & 2 & 0 & \infty \end{bmatrix}$$

Row Reduction :-

$$\begin{bmatrix} \infty & \infty & \infty & \infty \\ 0 & \infty & 6 & \infty \\ 0 & 3 & \infty & \infty \\ \infty & 2 & 0 & \infty \end{bmatrix}$$

Column Reduction :-

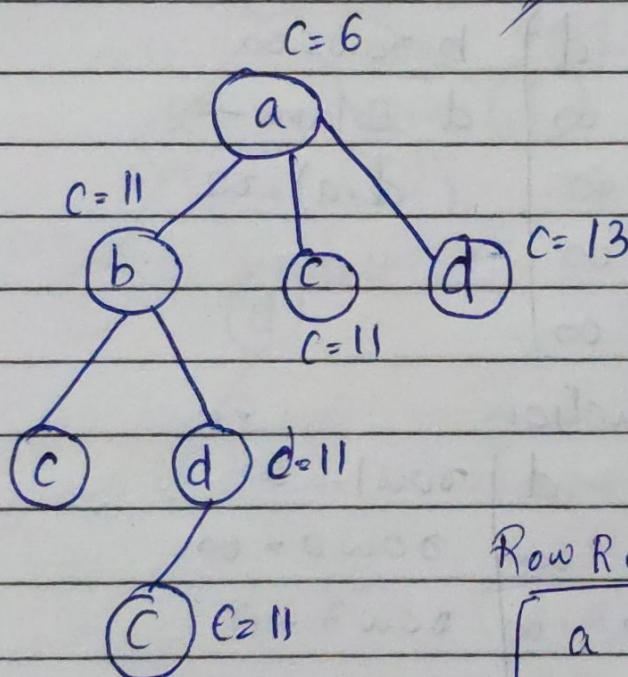
$$\begin{bmatrix} \infty & \infty & \infty & \infty \\ 0 & \infty & 6 & \infty \\ 0 & 1 & \infty & \infty \\ \infty & 0 & 0 & \infty \end{bmatrix}$$

no. of steps = $0 + 0 + 0 + 0 = 0$

no. of steps = $0 + 2 + 0 + 0 = 2$

$\gamma = 2 + 0 = 2$

$$\begin{aligned} \text{Cost} &= 6 + M_1(a, d) + \gamma \\ &= 6 + 5 + 2 \\ &= 13 \end{aligned}$$



B → C

$$\begin{bmatrix} \infty & \infty & \infty & \infty \\ \infty & \infty & \infty & \infty \\ \infty & \infty & \infty & \infty \\ 2 & \infty & \infty & \infty \end{bmatrix}$$

Row Reduction

$$\begin{array}{c|cccc} & a & b & c & d \\ \hline a & \infty & \infty & \infty & \infty \\ b & \infty & \infty & \infty & \infty \\ c & \infty & \infty & \infty & \infty \\ d & \infty & \infty & \infty & \infty \end{array}$$

Row 1 = ∞

Row 2 = ∞

Row 3 = ∞

Row 4 = 2

Column Reduction

	a	b	c	d	
a	∞	∞	∞	∞	Col 1 = 0
b	∞	∞	∞	∞	Col 2 = ∞ = 0
c	∞	∞	∞	∞	Col 3 = ∞ = 0
d	∞	∞	∞	∞	Col 4 = ∞ = 0

$$0 = 0 + 0 + 0 + 0 = 0 //$$

$$\text{Then Cost} = \text{Cost of Node 2} + M_2(b, c) + 0$$

$$= 11 + 5 + 2 = 18 //$$

(B, d)

	a	b	c	d	
a	∞	∞	∞	∞	b-row = ∞
b	∞	∞	∞	∞	d-Column = ∞
c	0	∞	∞	∞	(d, a) = ∞
d	∞	∞	0	∞	

Row Reduction

	a	b	c	d	
a	∞	∞	∞	∞	row 1 = ∞
b	∞	∞	∞	∞	row 2 = ∞
c	0	∞	∞	∞	row 3 = 0
d	∞	∞	0	∞	row 4 = 0

Column Reduction

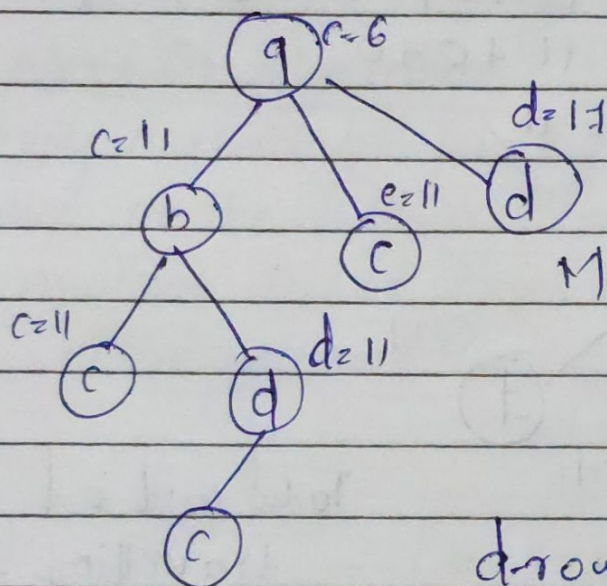
•

	a	b	c	d	col
a	∞	∞	∞	∞	col 2 = ∞
b	∞	∞	∞	∞	col 3 = 0
c	0	∞	∞	∞	col 4 = ∞
d	∞	∞	0	∞	

$$0 = 0$$

$$\text{Total Cost} = \text{Cost of Node } a + M_6(b, d) + r$$

$$= 11 + 0 + 0 = 11 //$$



$$M_6 =$$

	a	b	c	d
a	∞	∞	∞	∞
b	∞	∞	∞	∞
c	0	∞	∞	∞
d	∞	∞	0	∞

d-row
c-col
(c, a) = ∞

	a	b	c	d
a	∞	∞	∞	∞
b	∞	∞	∞	∞
c	∞	∞	∞	∞
d	∞	∞	∞	∞

Row Reduction and Column Reduction

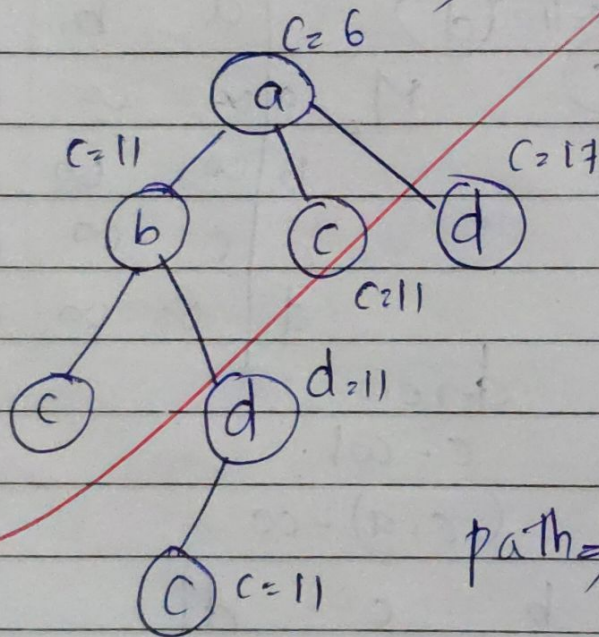
$$M_4 = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} \infty & \infty & \infty & \infty \\ \infty & \infty & \infty & \infty \\ \infty & \infty & \infty & \infty \\ \infty & \infty & \infty & \infty \end{bmatrix} \end{matrix}$$

$$0 = 0$$

$$\text{Total Cost} = \text{Cost of Node 6} + M(d, c) + 0$$

$$= 11 + 0 + 0$$

$$= 11 //$$



Total cost of travelling = 11 //

path $\Rightarrow a \rightarrow b \rightarrow d \rightarrow c \rightarrow a$

25/2/2025